Fair Cost Allocation in Energy Communities: A DLMP-based Bilevel Optimization with a Scalable Shapley Value Approach

Hyeongon Park, Kyuhyeong Kwag, Daniel K. Molzahn, and Rahul K. Gupta

***Abstract*—Energy communities (ECs) are emerging as a promising decentralized model for managing cooperative distributed energy resources (DERs). As these communities expand and their operations become increasingly integrated into the grid, ensuring fairness in allocating operating costs among participants becomes a challenge. Grid-induced operational constraints, such as the curtailment of solar generation within the EC, may affect members unequally. These disparities can lead to unfair cost distribution, particularly when Distribution Locational Marginal Prices (DLMPs) are employed. This highlights the need for cost-allocation mechanisms that account for both network constraints and participant-level impacts. Despite growing interest in fairness-aware methods, most methods do not account for the impact of DLMPs. To fill this gap, we propose a bilevel optimization model in which a Community Energy Aggregator (CEA) schedules DERs across multiple ECs while a Distribution System Operator (DSO) determines DLMPs through network-constrained dispatch. Leveraging the Karush-Kuhn-Tucker (KKT) conditions and strong duality, the bilevel model is reformulated into a tractable single-level problem. We achieve fairness in the cost allocation by applying the Shapley value to quantify each community's marginal contribution to system-wide cost savings. The effectiveness of the proposed method is validated through simulations on several benchmark distribution systems.**

***Index Terms*—Fairness-aware, Energy Community, LinDistFlow, DLMP, Shapley Value**

NOMENCLATURE

Sets

$\mathcal{N}$	Set of all distribution buses.
$\mathcal{N}_{\text{c}}$	Set of buses where communities are located.
$\mathcal{N}_s$	Set of slack buses (single bus at index 0).
$\mathcal{L}$	Set of distribution lines (l,k) in the network.
$\mathcal{T}$	Set of time periods in the scheduling horizon.

Parameters

$\hat{P}^{\text{pv}}_{l,t}$	Forecasted available PV generation.
$S^{\text{pv}}_{l,\text{max}}$	Apparent power rating of PV inverter.
PF_{min}	Minimum allowable power factor of PV inverter.
ζ^{pv}	Reactive power coefficient derived from PF_{min}.
$\overline{P}^{\text{ch}}_l$,$\overline{P}^{\text{dis}}_l$	Charging/discharging power capacities of BESS.
$\underline{E}_l$, $\overline{E}_l$	Min/max energy limits of BESS.
η^{ch}_l, η^{dis}_l	Charging/discharging efficiencies of BESS.
$\hat{P}^{\text{load}}_{l,t}$, $\hat{Q}^{\text{load}}_{l,t}$	Forecasted active/reactive load of community.
π	Compensation rate for curtailed load.
r_{kl}, x_{kl}	Resistance and reactance of line (k,l).
z_{kl}	Impedance of line (k,l), i.e., $z_{kl} = r_{kl} + jx_{kl}$.
π^{LMP}_t	Wholesale electricity price.
c^g_l	Marginal cost of local generator.
$\underline{p}^g_l$, $\overline{p}^g_l$	Active power output limits of generator.
$\underline{q}^g_l$, $\overline{q}^g_l$	Reactive power output limits of generator.
v^{min}, v^{max}	Lower/upper limits of bus voltage magnitude.

Decision Variables

$p^{\text{pv}}_{l,t}$, $q^{\text{pv}}_{l,t}$	Active /reactive power outputs of PV system.
$p^{\text{ch}}_{l,t}$, $p^{\text{dis}}_{l,t}$	Charging/discharging power of BESS.
$u^{\text{ch}}_{l,t}$, $u^{\text{dis}}_{l,t}$	Binary indicators for charging/discharging mode of BESS.
$e_{l,t}$	State of Charge of BESS.
$p^{\text{load}}_{l,t}$	Scheduled controllable load.
$p^c_{l,t}$, $q^c_{l,t}$	Active/reactive power consumption of the community.
$\lambda^p_{l,t}$, $\lambda^q_{l,t}$	DLMPs for active and reactive power.
$p^g_{l,t}$, $q^g_{l,t}$	Active/reactive power outputs of the generator.
$u_{l,t}$	Squared voltage magnitude.
$P_{lk,t}$, $Q_{lk,t}$	Active/reactive power flows from bus l to k.
$S_{lk,t}$	Complex power flow from bus l to k.
$f_{lk,t}$	Current magnitude on line (l,k).
$p^{\text{inj}}_{l,t}$, $q^{\text{inj}}_{l,t}$	Net active/reactive power injection.

Lagrange Multipliers

$\lambda^{\text{volt}}_{lk,t}$	Multiplier for voltage drop constraint on line (l,k).
$\underline{\lambda}^u_{l,t}$, $\overline{\lambda}^u_{l,t}$	Multipliers for lower and upper bounds on squared voltage magnitude.
$\lambda^p_{l,t}$, $\lambda^q_{l,t}$	Multipliers for active and reactive power balance.
$\underline{\lambda}^{gp}_{l,t}$, $\overline{\lambda}^{gp}_{l,t}$	Multipliers for lower and upper bounds on $p^g_{l,t}$.
$\underline{\lambda}^{gq}_{l,t}$, $\overline{\lambda}^{gq}_{l,t}$	Multipliers for lower and upper bounds on $q^g_{l,t}$.

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I. INTRODUCTION

The rapid proliferation of prosumers equipped with renewable energy resources (RES) and battery energy storage systems (BESS) has significantly increased interest in decentralized energy systems in recent years [1]. One such example is the rise of energy communities (ECs) [2], which coordinate distributed energy resources (DERs) and flexible loads at the local level. While ECs offer various technical and economic benefits, such as improved self-consumption, peak shaving, and reduced grid dependency, their effective formation and operation pose several challenges. A critical issue lies in the fair allocation of collective operation costs and benefits arising from shared assets or coordinated operation within the distribution grid [3].

In this context, recent studies have increasingly explored fair allocation mechanisms among participants. For example, [4] adopts a Nash bargaining-based priority mechanism for peer-to-peer energy trading in local communities, and a nucleolus analysis is used in [5] to ensure coalition stability in transmission expansion planning models. The Shapley value [6] has also attracted attention as an approach for fair cost and benefit allocation. This approach distributes the total cost (or benefit) according to each participant's marginal contribution across all possible coalitions and satisfies key fairness properties, including efficiency, symmetry, and the null player property [7]. These characteristics make it a transparent solution for cooperative cost sharing, particularly when multiple parties share infrastructure or coordinate operations.

Building on this idea, several studies have applied the Shapley value in the context of investment allocation for ECs. For instance, Rui et al. [8] proposed a Shapley-based incentive scheme to allocate the cost of grid-enhancing technologies fairly among multiple participants. Similarly, Pedrero et al. [9] introduced a scalable Shapley value method to determine fair investment strategies for large energy communities, focusing on equitable distribution of infrastructure costs such as shared batteries or RES installations. The Shapley value has also been incorporated into the design of market mechanisms [10], [11].

Moreover, in future distribution systems, bus-level electricity prices are proposed through Distribution Locational Marginal Prices (DLMPs) [12], which reflect the spatiotemporal value of energy and grid constraints. DLMP concepts have been widely studied as pricing signals but have not been applied to fair cost allocation among ECs. In practice, each EC's operation can influence DLMPs through its net injections and location. Thus, an EC's scheduling decisions affect both its own cost and system-wide DLMPs and cost distribution.

To show the impact of DLMP-based pricing on fairness, let us consider dispatch for a 3-bus toy-system in Fig. 1.The DSO minimizes the total generation cost subject to bus power balance and a voltage constraint:

$$\begin{aligned} \min_{p_0^g,\, p_2^g,\, P_{01},\, P_{12}} \quad & c_0 p_0^g + c_2 p_2^g \\ \text{s.t.} \quad & p_0^g - P_{01} = 0 && : \lambda_0 \\ & P_{01} - d_1 - P_{12} = 0 && : \lambda_1 \\ & P_{12} + p_2^g - (d_2 - \Delta) = 0 && : \lambda_2 \\ & u_2 \geq \underline{u} && : \mu \geq 0 \\ & 0 \leq p_2^g \leq \overline{p}_2^g, \end{aligned} \tag{1}$$

where $c_2 > c_0$ denotes the higher marginal cost of the local generator, which is dispatched by the DSO. Here, d_1 and d_2 are fixed demands, while $\Delta \geq 0$ represents a controllable net injection (e.g., battery discharge) provided by Community B.

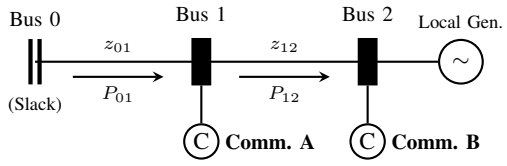

Fig. 1. The 3-bus radial distribution feeder used in the motivating example. A voltage lower bound is imposed only at the downstream bus (bus 2).

TABLE I
COST–BENEFIT ALLOCATION UNDER DLMP-BASED SETTLEMENT

	Community A (Bus 1)	Community B (Bus 2)
Action	None	Injection Δ
Injection cost	0	$c_\Delta \Delta$
DLMP change	$r_{01}\Delta\mu$	$(r_{01}+r_{12})\Delta\mu$
Share of total benefit†	$\frac{d_1 r_{01}}{d_1 r_{01} + d_2(r_{01}+r_{12})}$	$\frac{d_2(r_{01}+r_{12})}{d_1 r_{01} + d_2(r_{01}+r_{12})}$
System contribution	0	100%

† Benefit is defined as the reduction in DLMP-based payment, i.e., $d_i\,|\Delta\lambda_i|$.

Using the LinDistFlow approximation, the squared voltage magnitude at bus 2 is expressed as $u_2 = u_0 - 2\big[r_{01}d_1 + (r_{01} + r_{12})(d_2 - \Delta)\big]$. The voltage constraint $u_2 \geq \underline{u}$ can therefore be rewritten as $r_{01}d_1 + (r_{01} + r_{12})(d_2 - \Delta) \leq \frac{1}{2}(u_0 - \underline{u})$. Since $r_{01} + r_{12} > r_{01}$, injections at bus 2 are more effective in relieving the voltage constraint than injections at bus 1.

The DLMP at each bus is the Lagrange multiplier associated with the corresponding bus power balance constraint. From the KKT conditions of (1), the slack-bus price satisfies $\lambda_0 = c_0$, and the DLMPs are $\lambda_1 = \lambda_0 + r_{01}\mu$, $\lambda_2 = \lambda_0 + (r_{01} + r_{12})\mu$.

When the voltage constraint is inactive ($\mu = 0$), the prices are uniform, i.e., $\lambda_1 = \lambda_2 = c_0$. However, when the voltage constraint becomes binding ($\mu > 0$), the DSO must dispatch the more expensive local generator at bus 2, which raises the DLMPs at both buses, with a larger increase at bus 2 due to the higher sensitivity.

Now consider an increase in the injection Δ by Community B. This action relaxes the downstream voltage constraint, reducing the associated shadow price μ. As a result, the DLMPs at both buses decrease: $\Delta \uparrow \Rightarrow\ \mu \downarrow \Rightarrow\ \lambda_1 \downarrow,\ \lambda_2 \downarrow$.

This example reveals a cross-community externality under DLMP-based settlement. As shown in Table I, Community B bears the full cost of the corrective action, while the resulting DLMP reduction benefits both communities, including Community A, which does not change its operation. This highlights a fairness challenge for a CEA managing multiple communities and motivates the use of a contribution-based mechanism, such as the Shapley value.

The toy example above motivates the need for fairness-aware mechanisms; however, existing fairness-aware methods [3], [13]–[16] do not account for DLMP variations due to the ECs' coordination. These works assume fixed price environments or simplified distribution models.

Another challenge with the Shapely-based method are their high computational complexity for their application in large-scale systems [17]–[23]. To address this, approximation methods based on statistical sampling have been proposed, such as the stratified sampling highlighted in [17]. Others suggest grouping or clustering players to reduce the problem's dimensionality. However, the criteria for grouping in these studies are typically based on players' operational and economic characteristics, such as consumption patterns [18], their energy profiles [19], or market impact [20]. While these approaches reduce complexity, they do not directly account for the underlying physical characteristics of the power grid, which fundamentally determine the players' interactions.

In light of these gaps, this paper proposes a fair cost allocation framework for a Community Energy Aggregator (CEA)

coordinating multiple DER-equipped ECs accounting for the DLMP-based pricing. The problem is formulated as a bilevel optimization problem accounting for the interaction between the CEA and the Distribution System Operator (DSO). The scheme determines the DLMPs, which are then used to determine fair cost allocation using the Shapley value. To achieve scalability for Shapely-value method on large-scale systems, we propose a novel signature-based approximation scheme. The proposed framework can offer valuable policy insights into (i) fairness-aware community coordination, (ii) incentive-compatible market design, and (iii) mechanisms to encourage DER participation in distribution-level markets.

The key contributions of this work are listed below.

(i) We develop a **tractable single-level reformulation of the conventional bilevel DLMP-based scheme** using the Karush-Kuhn-Tucker (KKT) conditions and strong duality, enabling numerical solution under realistic settings.
(ii) We propose a **Shapley value-based fair cost allocation** method that redistributes each community's marginal contribution using DLMPs, and we evaluate its performance through case studies on benchmark distribution networks.
(iii) We propose a **signature-based approximation** method for **scalable Shapley value computation** based on their physical and topological characteristics.

The remainder of this paper is organized as follows. Section II describes the operational policies of the CEA and the DSO, including the problem structure and mathematical formulation. Section III presents the coalitional game-theoretic model and details the Shapley value method for fair cost allocation among communities. In Section IV, case studies are provided to evaluate the proposed methods, and Section V summarizes the findings and concludes the paper.

II. CEA & DSO Scheduling Framework

A. Description of Problem Structure

Our goal is to coordinate the operation of different local communities and the distribution network. This is modeled using a bilevel optimization framework, where the CEA and the DSO are structured as the leader and the follower, respectively. The overall structure of this scheme is illustrated in Fig. 2. The upper-level problem represents the CEA's decision-making on community-level energy scheduling, while the lower-level problem reflects the DSO's response in maintaining system-wide balance and operational feasibility.

The CEA optimizes the operation of the ECs connected to the distribution grid, while the DSO is responsible for the reliable operation of the distribution network. We assume that each EC hosts various DERs (e.g., photovoltaic (PV), BESS, and controllable loads) that are directly controlled and optimized by the CEA. The DSO procures electricity at the transmission-distribution interface at wholesale market prices, supplies power to both non-community consumers and ECs, and controls local generators while respecting distribution network constraints.

The price signals used in CEA scheduling are DLMPs derived from the DSO's optimization problem. These DLMPs are the dual variables associated with active and reactive power

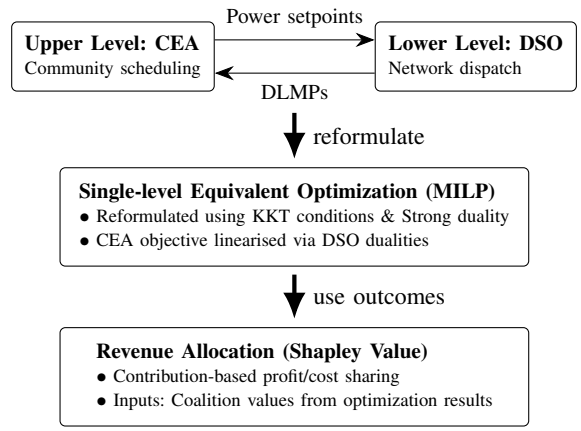

Fig. 2. Proposed framework: Bilevel interaction, single-level reformulation, and Shapley-based revenue allocation.

balance constraints at each bus. Consequently, each CEA's cost is affected by the locational and temporal conditions of the distribution grid. At the same time, the CEAs' scheduling decisions influence the bus power injections affecting the DSO's power balance and the resulting DLMPs. This interdependency results in a coupled bilevel optimization problem. We next describe the CEA and DSO models.

B. CEA Model: Upper Level Problem

The objective of the CEA is to minimize the total cost of active and reactive power consumption under DLMP-based price signals (from DSO), along with the incentive payments for flexible load reductions. The CEA optimization problem is

$$\min \sum_{t\in\mathcal{T}}\sum_{l\in\mathcal{N}_c}\left(\lambda^p_{l,t}\cdot p^c_{l,t}+\lambda^q_{l,t}\cdot q^c_{l,t}+\pi\cdot(\hat{P}^{\text{load}}_{l,t}-p^{\text{load}}_{l,t})\right) \tag{2}$$

$$\text{subject to: } 0\le p^{\text{pv}}_{l,t}\le \hat{P}^{\text{pv}}_{l,t} \tag{3a}$$

$$(p^{\text{pv}}_{l,t})^2+(q^{\text{pv}}_{l,t})^2\le(S^{\text{pv}}_{l,\max})^2 \tag{3b}$$

$$q^{\text{pv}}_{l,t}\le p^{\text{pv}}_{l,t}\zeta^{\text{pv}};\quad -q^{\text{pv}}_{l,t}\le p^{\text{pv}}_{l,t}\zeta^{\text{pv}} \tag{3c}$$

$$0\le p^{\text{ch}}_{l,t}\le \overline{P}^{\text{ch}}_l\cdot u^{\text{ch}}_{l,t}\ 0\le p^{\text{dis}}_{l,t}\le \overline{P}^{\text{dis}}_l\cdot u^{\text{dis}}_{l,t} \tag{3d}$$

$$u^{\text{ch}}_{l,t}+u^{\text{dis}}_{l,t}\le 1 \tag{3e}$$

$$e_{l,t+1}=e_{l,t}+\eta^{\text{ch}}_l p^{\text{ch}}_{l,t}\Delta t-(1/\eta^{\text{dis}}_l)p^{\text{dis}}_{l,t}\Delta t \tag{3f}$$

$$\underline{E}_l\le e_{l,t}\le \overline{E}_l \tag{3g}$$

$$0\le p^{\text{load}}_{l,t}\le \hat{P}^{\text{load}}_{l,t} \tag{3h}$$

$$p^c_{l,t}=p^{\text{load}}_{l,t}-p^{\text{pv}}_{l,t}-p^{\text{dis}}_{l,t}+p^{\text{ch}}_{l,t} \tag{3i}$$

$$q^c_{l,t}=\hat{Q}^{\text{load}}_{l,t}-q^{\text{pv}}_{l,t} \tag{3j}$$

Here, $\lambda^p_{l,t}$ and $\lambda^q_{l,t}$ represent DLMPs for active and reactive power and are treated as input parameters in the CEA problem. They are, however, determined as dual variables associated with power balance constraints in the DSO model described later in Section II-C. It is important to note that the CEA only considers community buses in its objective function and excludes the operation of other non-community buses.

The CEA scheduling problem includes constraints on DER operations across all community buses $l\in\mathcal{N}_c$ and time periods $t\in\mathcal{T}$. These constraints consist of (i) PV generation limits, (ii) BESS operating rules, (iii) controllable load bounds, and (iv) the resulting net power consumptions submitted to the DSO. The complete set of constraints is outlined as follows.

The PV operating limits are defined by (3a)–(3c), including real power availability, inverter capacity, and power factor requirements. The BESS model, given in (3d)–(3g), captures charging and discharging limits, state-of-charge dynamics, and mutually exclusive operating modes. Controllable loads are bounded within their flexibility limits as specified in (3h). The net active and reactive power consumption at each community bus is determined by (3i) and (3j). The net injection reflects the scheduled load minus local DER outputs, including PV generation and BESS discharging (or charging). These net injections are submitted to the DSO for network-constrained dispatch and DLMP calculation. In this work, BESS units are assumed not to provide reactive power support.

C. DSO Model: Lower Level Problem

The DSO aims to ensure the reliable and cost-effective operation of the distribution grid in response to the power injections scheduled by the CEAs. Specifically, it minimizes the total procurement and dispatch costs while maintaining system-wide power balance and voltage limits. The upstream power purchase is settled at the wholesale market price (π_t^{LMP}), while dispatchable local generators are operated based on their marginal costs. In this model, the DSO determines the DLMPs through the dual variables of the power balance constraints, which are used by CEAs as price signals for subsequent scheduling. The DSO optimization problem is

$$\min \sum_{t\in\mathcal{T}} \pi_t^{\text{LMP}} \cdot p_{0,t}^g + \sum_{t\in\mathcal{T}} \sum_{l\in\mathcal{N}\setminus\mathcal{N}_s} c_l^g \cdot p_{l,t}^g \tag{4}$$

subject to:

$$u_{k,t} = u_{l,t} - 2(r_{kl}P_{lk,t} + x_{kl}Q_{lk,t}) \quad : \lambda_{lk,t}^{\text{volt}} \tag{5a}$$

$$(v^{\min})^2 \le u_{l,t} \le (v^{\max})^2 \quad : \underline{\lambda}_{l,t}^u, \overline{\lambda}_{l,t}^u \tag{5b}$$

$$\sum_{k:(0,k)\in\mathcal{L}} P_{0k,t} - p_{0,t}^{\text{inj}} - p_{0,t}^g = 0 \quad : \lambda_{0,t}^p \tag{5c}$$

$$\sum_{k:(l,k)\in\mathcal{L}} P_{lk,t} - \sum_{k:(k,l)\in\mathcal{L}} P_{kl,t} - p_{l,t}^{\text{inj}} - p_{l,t}^g = 0 : \lambda_{l,t}^p \tag{5d}$$

$$\sum_{k:(0,k)\in\mathcal{L}} Q_{0k,t} - q_{0,t}^{\text{inj}} - q_{0,t}^g = 0 \quad : \lambda_{0,t}^q \tag{5e}$$

$$\sum_{k:(l,k)\in\mathcal{L}} Q_{lk,t} - \sum_{k:(k,l)\in\mathcal{L}} Q_{kl,t} - q_{l,t}^{\text{inj}} - q_{l,t}^g = 0 : \lambda_{l,t}^q \tag{5f}$$

$$\underline{p}_l^g \le p_{l,t}^g \le \overline{p}_l^g \quad : \underline{\lambda}_{l,t}^{gp},\ \overline{\lambda}_{l,t}^{gp} \tag{5g}$$

$$\underline{q}_l^g \le q_{l,t}^g \le \overline{q}_l^g \quad : \underline{\lambda}_{l,t}^{gq},\ \overline{\lambda}_{l,t}^{gq} \tag{5h}$$

Here, (5a)–(5f) represent the power-flow model using the linearized DistFlow equations, ensuring a tractable convex formulation [24]. In this approximation, quadratic power loss terms are neglected. Eqs. (5g) and (5h) limit the generators' active and reactive power outputs. Note that these constraints are defined for all buses $l \in \mathcal{N}$ and time periods $t \in \mathcal{T}$, although such indices are omitted in individual expressions for brevity. The resulting linearized constraints and their associated Lagrange multipliers $\lambda^\bullet$ are shown next to the constraints. In the power balance constraint (5d), power injections are related to the net active and reactive power consumption of the connected community and load forecasts as $p_{l,t}^{\text{inj}} = -p_{l,t}^c, \forall l \in \mathcal{N}_c$ otherwise $p_{l,t}^{\text{inj}} = -\hat{P}_l^{\text{load}}, \forall l \in \mathcal{N}\setminus\mathcal{N}_c$. Similarly, reactive power injections follow $q_{l,t}^{\text{inj}} = -q_{l,t}^c,\ \forall l \in \mathcal{N}_c$ and $q_{l,t}^{\text{inj}} = -\hat{Q}_l^{\text{load}},\ \forall l \in \mathcal{N}\setminus\mathcal{N}_c$.

D. Single-level Reformulation via KKT Conditions

As described in Section II-A, the interaction between the CEA and the DSO is formulated as a bilevel optimization problem, where the CEA (upper-level) determines community-level energy schedules, and the DSO (lower-level) responds with optimal dispatch and pricing decisions. To obtain a tractable formulation, the lower-level DSO problem is replaced by its KKT optimality conditions [25]. We start by writing the Lagrangian using the DSO objective and constraints (5) as

$$\begin{aligned}
\mathcal{L} = &\sum_{t\in\mathcal{T}} \pi_t^{\text{LMP}} \cdot p_{0,t}^g + \sum_{t\in\mathcal{T}} \sum_{l\in\mathcal{N}\setminus\mathcal{N}_s} c_l^g \cdot p_{l,t}^g \\
&+ \sum_{t\in\mathcal{T}} \sum_{(l,k)\in\mathcal{L}} \lambda_{lk,t}^{\text{volt}} \left(u_{k,t} - u_{l,t} + 2(r_{kl}P_{lk,t} + x_{kl}Q_{lk,t})\right) \\
&+ \sum_{t\in\mathcal{T}} \sum_{l\in\mathcal{N}} \underline{\lambda}_{l,t}^u \left((v^{\min})^2 - u_{l,t}\right) + \overline{\lambda}_{l,t}^u \Big(u_{l,t} - (v^{\max})^2\Big) \\
&+ \sum_{t\in\mathcal{T}} \lambda_{0,t}^p \Big(\sum_{k:(0,k)\in\mathcal{L}} P_{0k,t} - p_{0,t}^{\text{inj}} - p_{0,t}^g \Big) \\
&+ \sum_{t\in\mathcal{T}} \sum_{l\in\mathcal{N}\setminus\mathcal{N}_s} \lambda_{l,t}^p \Big(\sum_{k:(l,k)\in\mathcal{L}} P_{lk,t} - \sum_{k:(k,l)\in\mathcal{L}} P_{kl,t} - p_{l,t}^{\text{inj}} - p_{l,t}^g \Big) \\
&+ \sum_{t\in\mathcal{T}} \lambda_{0,t}^q \Big(\sum_{k:(0,k)\in\mathcal{L}} Q_{0k,t} - q_{0,t}^{\text{inj}} - q_{0,t}^g \Big) \\
&+ \sum_{t\in\mathcal{T}} \sum_{l\in\mathcal{N}\setminus\mathcal{N}_s} \lambda_{l,t}^q \Big(\sum_{k:(l,k)\in\mathcal{L}} Q_{lk,t} - \sum_{k:(k,l)\in\mathcal{L}} Q_{kl,t} - q_{l,t}^{\text{inj}} - q_{l,t}^g \Big) \\
&+ \sum_{t\in\mathcal{T}} \sum_{l\in\mathcal{N}} \left[\underline{\lambda}_{l,t}^{gp} \left(\underline{p}_l^g - p_{l,t}^g\right) + \overline{\lambda}_{l,t}^{gp} \left(p_{l,t}^g - \overline{p}_l^g\right) \right] \\
&+ \sum_{t\in\mathcal{T}} \sum_{l\in\mathcal{N}} \left[\underline{\lambda}_{l,t}^{gq} \left(\underline{q}_l^g - q_{l,t}^g\right) + \overline{\lambda}_{l,t}^{gq} \left(q_{l,t}^g - \overline{q}_l^g\right) \right]
\end{aligned} \tag{6}$$

Now, we can write the DSO problem's KKT conditions, consisting of four components: primal feasibility, stationarity, dual feasibility, and complementarity. Primal feasibility requires satisfying all DSO constraints, including the LinDistFlow equations and operational limits defined in (5).

Let $x_{\text{DSO}} := \{P_{lk,t}, Q_{lk,t}, u_{l,t}, p_{l,t}^g, q_{l,t}^g\}$ denote the set of DSO primal variables. Stationarity requires that the gradient of the Lagrangian with respect to each element of x_{DSO} vanishes:

$$\frac{\partial \mathcal{L}}{\partial P_{lk,t}} = 2r_{kl}\lambda_{lk,t}^{\text{volt}} + \lambda_{l,t}^p - \lambda_{k,t}^p = 0 \tag{7a}$$

$$\frac{\partial \mathcal{L}}{\partial Q_{lk,t}} = 2x_{kl}\lambda_{lk,t}^{\text{volt}} + \lambda_{l,t}^q - \lambda_{k,t}^q = 0 \tag{7b}$$

$$\frac{\partial \mathcal{L}}{\partial u_{l,t}} = -\sum_{k:(l,k)\in\mathcal{L}} \lambda_{lk,t}^{\text{volt}} + \sum_{k:(k,l)\in\mathcal{L}} \lambda_{kl,t}^{\text{volt}} - \underline{\lambda}_{l,t}^u + \overline{\lambda}_{l,t}^u = 0 \tag{7c}$$

$$\frac{\partial \mathcal{L}}{\partial p_{0,t}^g} = \pi_t^{\text{LMP}} - \lambda_{0,t}^p - \underline{\lambda}_{0,t}^{gp} + \overline{\lambda}_{0,t}^{gp} = 0 \tag{7d}$$

$$\frac{\partial \mathcal{L}}{\partial q_{0,t}^g} = -\lambda_{0,t}^q - \underline{\lambda}_{0,t}^{gq} + \overline{\lambda}_{0,t}^{gq} = 0 \tag{7e}$$

$$\frac{\partial \mathcal{L}}{\partial p_{l,t}^g} = c_l^g - \lambda_{l,t}^p - \underline{\lambda}_{l,t}^{gp} + \overline{\lambda}_{l,t}^{gp} = 0 \tag{7f}$$

$$\frac{\partial \mathcal{L}}{\partial q^g_{l,t}} = -\lambda^q_{l,t} - \underline{\lambda}^{gq}_{l,t} + \overline{\lambda}^{gq}_{l,t} = 0. \tag{7g}$$

Let $\mathcal{I}$ denote the index set of all DSO inequality constraints, including voltage magnitude limits and active/reactive generation capacity limits, each written as $g_i(x_{\text{DSO}}) \leq 0$ with associated dual variable λ_i. Dual feasibility requires

$$\lambda_i \geq 0, \qquad \forall i \in \mathcal{I}. \tag{8}$$

Complementarity conditions between each inequality constraint and its corresponding dual variable are enforced using a standard Big-Mreformulation [26]. For each $i \in \mathcal{I}$, a binary variable $z_i \in \{0, 1\}$ is introduced, yielding

$$0 \leq \lambda_i \leq \mathfrak{M} z_i, \tag{9a}$$

$$g_i(x_{\text{DSO}}) \leq \mathfrak{M}(1 - z_i), \qquad \forall i \in \mathcal{I}. \tag{9b}$$

The conditions above ensure that any feasible solution of the dual problem corresponds to an optimal solution of the primal DSO problem. The original CEA objective (2) contains the DLMP values $\lambda^p_{l,t} p^c_{l,t}$ and $\lambda^q_{l,t} q^c_{l,t}$. After replacing the lower-level DSO problem by its KKT conditions, the multipliers $\lambda^p_{l,t}$ and $\lambda^q_{l,t}$ become decision variables of the single-level problem, which makes it bilinear in (λ, p^c, q^c). Since the DSO problem is linear and feasible, strong duality holds. Thus, the primal and dual DSO objectives are equal. Using this property, the bilinear terms in (2) are replaced by their dual representation, yielding the linear objective in (10). The derivation is provided in Appendix A.

$$\begin{aligned} f^{\text{SINGLE}} = & \sum_{t \in \mathcal{T}} \pi^{\text{LMP}}_t p^g_{0,t} + \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{N} \setminus \mathcal{N}_s} c^g_l p^g_{l,t} \\ & - \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{N} \setminus \mathcal{N}_c} \lambda^p_{l,t} \hat{P}^{\text{load}}_{l,t} - \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{N} \setminus \mathcal{N}_c} \lambda^q_{l,t} \hat{Q}^{\text{load}}_{l,t} \\ & + \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{N}_c} \pi(\hat{P}^{\text{load}}_{l,t} - p^{\text{load}}_{l,t}) \\ & + \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{N}} \left(\overline{\lambda}^{gp}_{l,t} \overline{p}^g_l - \underline{\lambda}^{gp}_{l,t} \underline{p}^g_l \right) + \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{N}} \left(\overline{\lambda}^{gq}_{l,t} \overline{q}^g_l - \underline{\lambda}^{gq}_{l,t} \underline{q}^g_l \right) \\ & + \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{N}} \left(\overline{\lambda}^u_{l,t} (v^{\max})^2 - \underline{\lambda}^u_{l,t} (v^{\min})^2 \right). \end{aligned} \tag{10}$$

Note that all terms in the objective function are linear as a result of the dual substitution. Finally, the original bilevel CEA–DSO problem is reformulated as the following single-level mixed-integer linear program (MILP):

$$\begin{aligned} \min \quad & f^{\text{SINGLE}} \ (10) \\ \text{subject to:} \quad & \text{CEA constraints: } (3), \\ & \text{DSO KKT conditions: } (5), (7), (8), (9) \end{aligned} \tag{11}$$

This reformulation removes the nested structure of the bilevel problem by embedding the DSO optimality conditions into the upper-level model, as illustrated in Fig. 2.

III. Fair Cost Allocation using Coalitional Game and Shapley Value Method

The optimal value obtained by (11) optimizes the total operational cost aggregated over all participating ECs. Although this cost is optimal from the CEA's perspective, it does not guarantee fairness in allocating costs to individual ECs due to the grid condition induced DLMPs. This necessitates a formal fairness-based allocation mechanism. In this section, we present an approach for fair cost allocation using the Shapely Value method [27]. Using this approach, fair cost allocation is treated as a coalitional game to distribute cooperative savings based on marginal contributions of each ECs.

Let $\mathcal{M}$ denote the set of participating ECs, and let $\mathcal{C} \subseteq \mathcal{M}$ represent a coalition. The characteristic function $\upsilon : 2^{\mathcal{M}} \to \mathbb{R}$ assigns to each coalition $\mathcal{C}$ the corresponding cooperative cost savings. The individual cost c^{indiv}_m of each EC $m \in \mathcal{M}$ is obtained by solving (11) with only EC m participating, while all other ECs, i.e., those in $\mathcal{M} \setminus \{m\}$, operate under their baseline (uncontrolled) profiles.

Similarly, for a coalition $\mathcal{C} \subseteq \mathcal{M}$, the cooperative cost $c_{\mathcal{C}}$ is computed by jointly optimizing the ECs in $\mathcal{C}$, whereas the non-participating ECs, i.e., those in $\mathcal{M} \setminus \mathcal{C}$, remain under their baseline profiles. The value of a coalition $\mathcal{C}$ is then be defined as the total cost savings achieved by the coalition relative to operating individually:

$$\upsilon(\mathcal{C}) = \sum_{m \in \mathcal{C}} c^{\text{indiv}}_m - c_{\mathcal{C}}. \tag{12}$$

Using the Shapley value method [6], for a set of communities $\mathcal{M}$ with $|\mathcal{M}| = N$, the allocation $\phi_m(\upsilon)$ to community m is defined as:

$$\phi_m(\upsilon) = \sum_{\mathcal{S} \subseteq \mathcal{M} \setminus m} \frac{|\mathcal{S}|!(N - |\mathcal{S}| - 1)!}{N!} [\upsilon(\mathcal{S} \cup m) - \upsilon(\mathcal{S})], \tag{13}$$

where $\upsilon(\mathcal{S} \cup \{m\}) - \upsilon(\mathcal{S})$ represents the marginal contribution of community m to a coalition $\mathcal{S}$. This allocation satisfies the efficiency property, $\sum_{m \in \mathcal{M}} \phi_m(\upsilon) = \upsilon(\mathcal{M})$, ensuring that the total savings from joint optimization are fully and fairly distributed based each community contribution.

Note that the optimization problems associated with different coalitions are independent and can therefore be solved in parallel. This enables a substantial reduction in computation time for Shapley value evaluation. However, despite parallelization, the total number of coalitions grows exponentially with the number of ECs, which limits the scalability of full enumeration for large systems. To further mitigate this issue, we propose a signature-based approximation method that efficiently samples a representative subset of coalitions. The computational performance of this framework, including the synergy between parallelization and approximation, is systematically evaluated in Section IV.

The final cost for each community is obtained by subtracting its share of the cost savings from its individual cost:

$$c^{\text{final}}_m = c^{\text{indiv}}_m - \phi_m(\upsilon) \tag{14}$$

IV. Numerical Validation

A. Simulation Setup

We evaluate the effectiveness of the proposed fair allocation scheme by numerical simulations on various realistic distribution networks. First, we present results for the CIGRE low-voltage system, then demonstrate scalability by simulating it for case69 [24] and case123 [28].

Across all test cases, the same modeling assumptions and DER parameters are applied. Each energy community is composed of PV plants, BESS, and controllable loads. We assume the loads can be controlled up to 30% of their base demand through flexible load response, with a compensation rate of \$75/MWh [29]. The electricity price at the substation uses the 2024 average of hourly CAISO locational marginal prices (LMP) for the NP-15 zone, which varies between \$12/MWh and \$55/MWh [30]. Local dispatchable generators are assumed to be available at a few distribution buses, operating at a marginal cost of \$250/MWh, based on values reported in [31]. The reactive power capabilities of dispatchable generators and PV inverters are constrained by typical grid code requirements. Generator output is limited by a minimum power factor of 0.9, while PV inverters are restricted to supplying reactive power up to 20% of their real power output [32]. The charging and discharging efficiencies of BESS are both set to 95%, i.e., $\eta_l^{\text{ch}} = \eta_l^{\text{dis}} = 0.95$, and simultaneous charging and discharging are not permitted. The initial and final state of charge (SoC) is set to 50% of the rated capacity. We model the load and PV generation using real measurements from the smart grid setup at the EPFL, Switzerland [33]. The data,[1] which is available at per second resolution, is down-sampled to hourly resolution.

For each community, we show four quantities: (i) *individual cost* c_m^{indiv} is obtained by solving (11) with only one particular community active while others are uncontrolled and uncoordinated. This represents the cost if each community operates independently without coordination, (ii) The *Shapley-based saving* $\phi_m(\upsilon)$ quantifies the marginal contribution of community m to the grand coalition (all ECs are jointly optimized for their coordinated operation), (iii) the *Final cost (Shapely)* settlement for each community is calculated as $c_m^{\text{final}} = c_m^{\text{indiv}} - \phi_m(\upsilon)$ according to (14), ensuring that the cost reduction from cooperation is allocated based on the contribution of each participant, and finally (iv) the *Final cost (Base)* cost, c_m^{base} where the cost is directly calculated from the DLMPs and dispatch from the solution to (11), without accounting for the community's marginal cost savings. We use Yalmip to implement the optimization problem, and Gurobi as solver. Simulations are done in MATLAB on a laptop with 24 GB of memory.

B. Simulation Results

1) CIGRE 19-bus system

Fig. 3 shows the benchmark CIGRE low-voltage distribution network used in this study [34]. The feeder is connected to the transmission system via a 20/0.4 kV, 400 kVA transformer. The system includes both community and non-community loads, and the impact of community locations on system-level fairness and grid operation is analyzed for multiple scenarios.

In the first scenario, we consider three communities located at buses 9, 11, and 18. Each community is initially assumed to have an identical DER setup consisting of PV, controllable demand, and a BESS rated at 20 kW/50 kWh. Fig. 4 compares the individual costs, the Shapley-based savings, and the final allocated costs under the Shapley and Base methods. We

[1]https://github.com/DESL-EPFL/DESL-Photovoltaic-timeseries

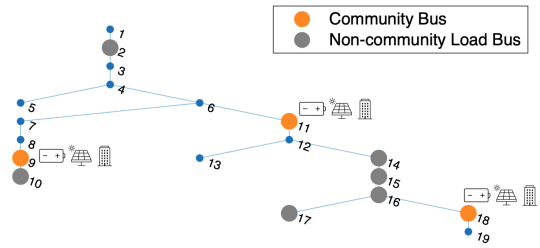

Fig. 3. CIGRE low voltage benchmark network. Local dispatchable generators are assumed to be available at buses 13, 16, and 18.

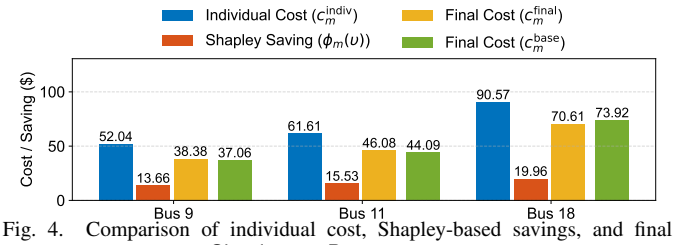

Fig. 4. Comparison of individual cost, Shapley-based savings, and final allocated cost under the Shapley and Base methods for communities at buses 9, 11, and 18.

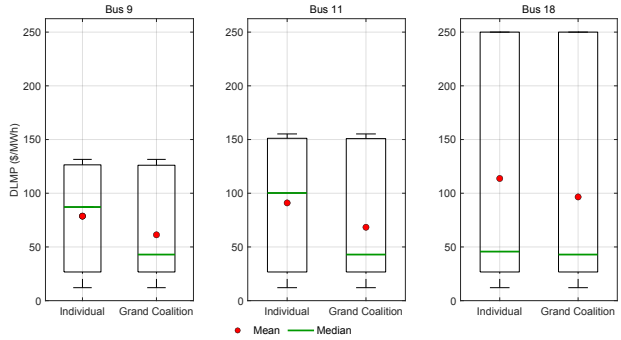

Fig. 5. Active power DLMP distributions at the community buses under individual participation and grand coalition participation. In the individual case, only one EC is optimized while the others follow baseline (uncontrolled) profiles; in the grand coalition case, all ECs in $\mathcal{M}$ are jointly optimized.

observe that communities under cooperation, i.e., in Final cost with Shapely and Base, the costs are lower compared to the without any cooperation (individual). Moreover, the Shapley method distributes the savings fairly based on marginal contributions, whereas the Base method directly uses DLMP results without considering individual contributions, which is lower than individual costs, but maybe unfair.

Among the three, the community located at bus 18 exhibits the highest individual and final costs, along with the largest Shapley value. This result can be further explained by distributions of DLMPs for individual and cooperative case, shown in Fig. 5. As shown, the DLMPs at bus 18 are significantly higher, which might be due to binding voltage constraints. As a result, its contribution to reducing overall system costs is the most significant, which is reflected in its higher Shapley value. Across all buses, DLMP distributions shift lower under the grand coalition, indicating that coordinated participation benefits all communities.

Sensitivity with BESS size: As DLMP values might be affected due to BESS size, which then influences fairness, we consider scenarios in which the BESS capacity at a single site is selectively increased. Usually, the additional BESS enhances operational flexibility and enables peak shaving, leading to reduced system-wide costs. However, how these savings are

TABLE II
FINAL COST COMPARISON UNDER BESS EXPANSION (3× ORIGINAL CAPACITY) AT A SINGLE COMMUNITY. "COMM." DENOTES THE BUS ID OF EACH COMMUNITY.

Method	Scenario	Comm.	BESS x1 Cost ($)	BESS x3 Cost ($)	Cost Diff. ($)
Shapley	BESS@9 x3	**Bus 9**	**38.38**	**20.90**	**17.48**
		Bus 11	46.08	44.37	1.71
		Bus 18	70.61	68.31	2.30
	Total		**155.07**	**133.58**	**21.49**
	BESS@11 x3	Bus 9	38.38	36.99	1.39
		Bus 11	**46.08**	**26.28**	**19.80**
		Bus 18	70.61	68.91	1.70
	Total		**155.07**	**132.19**	**22.89**
	BESS@18 x3	Bus 9	38.38	38.81	-0.43
		Bus 11	46.08	43.24	2.84
		Bus 18	**70.61**	**39.30**	**31.31**
	Total		**155.07**	**121.36**	**33.72**
Base	BESS@9 x3	**Bus 9**	**37.06**	**26.44**	**10.62**
		Bus 11	44.09	41.09	3.00
		Bus 18	73.92	66.06	7.87
	Total		**155.07**	**133.58**	**21.49**
	BESS@11 x3	Bus 9	37.06	33.47	3.59
		Bus 11	**44.09**	**32.80**	**11.29**
		Bus 18	73.92	65.91	8.01
	Total		**155.07**	**132.19**	**22.89**
	BESS@18 x3	Bus 9	37.06	29.09	7.96
		Bus 11	44.09	35.96	8.14
		Bus 18	**73.92**	**56.31**	**17.62**
	Total		**155.07**	**121.36**	**33.72**

TABLE III
FINAL COST COMPARISON WHEN EXPANDING BESS AT BUS 11 TO 3.5× ITS ORIGINAL CAPACITY.

Method	Scenario	Comm.	BESS x1 Cost ($)	BESS x3.5 Cost ($)	Cost Diff. ($)
Shapley	BESS@11 x3.5	Bus 9	38.38	35.30	3.08
		Bus 11	**46.08**	**20.90**	**25.18**
		Bus 18	70.61	67.76	2.86
	Total		**155.07**	**123.95**	**31.12**
Base	BESS@11 x3.5	Bus 9	37.06	30.67	6.38
		Bus 11	**44.09**	**32.85**	**11.24**
		Bus 18	73.92	60.42	13.50
	Total		**155.07**	**123.95**	**31.12**

distributed across communities differs significantly depending on the allocation rule.

Table II presents the final Shapely and Base cost of each community when the BESS at one site is increased to three times its original capacity.The Shapley allocation more accurately attributes cost savings to the community providing additional BESS capacity. When bus 11 hosts the expanded BESS, its cost decreases from $46.08 to $26.28 (a $19.80 reduction). In contrast, under the Base allocation, bus 11's cost decreases by only $11.29 (from $44.09 to $32.80), while bus 18, despite contributing no additional resources, also benefits from a $8.01 reduction. This illustrates that the baseline method unfairly benefits non-contributing communities and does not fully capture localized contributions.

Table III further extends the analysis by expanding BESS capacity at bus 11 to 3.5 times its original size. Under the Shapley method, the total cost decreases from $155.07 to $123.95, with the largest benefit concentrated at bus 11, whose cost drops from $46.08 to $20.90 ($25.18 reduction). In contrast, under the Base method, the cost at bus 11 decreases from $44.09 to $32.85 ($11.24 reduction), while bus 18, without any new investment, sees its cost drop from $73.92 to $60.42 ($13.50 reduction), which is even larger. This result highlights a potential limitation of the Base allocation: it may not fully reflect the origin of the cost savings, which could result in a less fair allocation. Such misalignment between contribution and benefit may weaken the incentive for communities to invest in grid-supportive resources like BESS.

In one of the cases, we encounter negative cost difference (bus 9 under the "BESS@18 x3" scenario) where the allocated cost under the Shapley method is slightly higher than the original cost. However, the increase is marginal, and the cost still remains lower than the individual cost (Fig. 4). Therefore, cooperation is still beneficial for all communities, and the individual rationality condition is satisfied.

Cost of non-cooperative communities with the number of cooperative communities: Here, we analyze the impact on the cost incurred by the non-participating communities with respect to the number of communities that participate in coalition. For this, we consider buses 3, 7, 9, 12, and 18 as cooperative communities, and we report the costs of non-participating buses 2, 10, 14, 15, 16, and 17 in Fig. 6. This figure shows the cost distribution for non-participating buses with an increasing number of participating communities. We show the results for buses 2, 14, 15, and 17, as others have similar trends. The results demonstrate that the average cost of non-community load buses decreases as more communities join the coalition due to overall lower DLMPs. However, even with the same number of participating communities, noticeable variation in costs is observed depending on the specific combination of communities in the coalition. This indicates that increasing the number of participating communities does not necessarily guarantee benefits for all non-participating buses, as the specific coalition composition can lead to widely varying outcomes.

2) IEEE 69-bus system

We extend the analysis to the IEEE 69-bus test system. We consider multiple scenarios depicting different placements

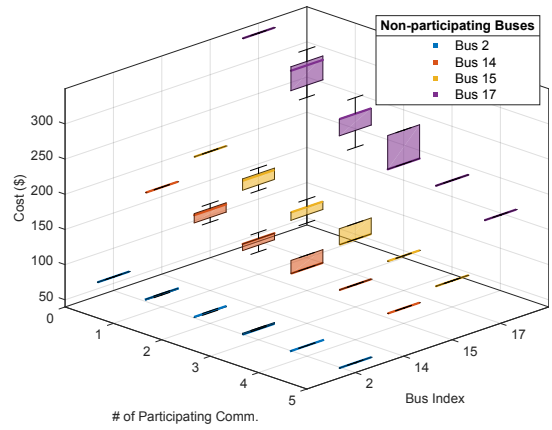

Fig. 6. 3D box-plot of non-participating buses' cost distribution across different numbers of participating communities.

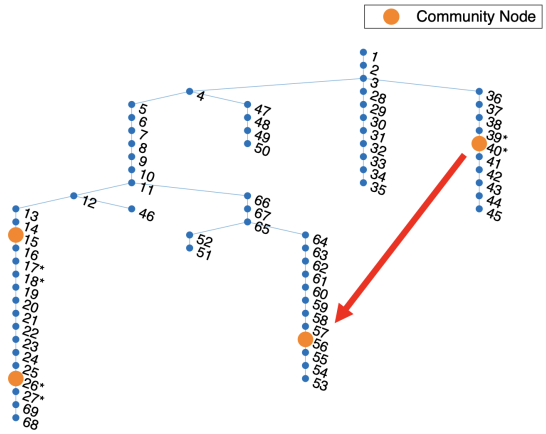

Fig. 7. IEEE 69-bus distribution system showing three different EC placement configurations. The first two configurations consider three ECs located at buses 15, 26, and 40 and at buses 15, 26, and 56, respectively. The third configuration places six ECs at buses 17, 18, 26, 27, 39, and 40, marked with an asterisk (∗) to demonstrate the signature-based approximation. Non-community load buses are omitted for clarity.

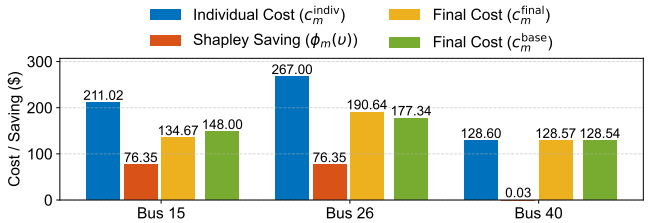

Fig. 8. Individual cost, Shapley-based savings, and final cost for each community bus in the IEEE 69-bus system (ECs at buses 15, 26, 40), under Shapley and Base allocation methods.

of ECs to examine the impact of community location on cost savings. All ECs are assumed to have identical DER portfolios and capacities. Again, we compare Shapley and Base methods as defined earlier.

Case 1 - ECs at buses 15, 26 and 40: We first examine a configuration where one EC is placed farther away (i.e., bus 40) from the others (i.e., buses 15 and 26). The results are visualized in Fig. 8. As observed, the cost for EC at bus 40 does not change much from the cooperation. This may be due to weak coupling between the EC's operations as they are far apart. In contrast, the ECs at buses 15 and 26 exhibit equal Shapley savings, reflecting their symmetric contributions to system-wide cost savings. This symmetry is not seen under the Base method, which leads to a relatively more favorable allocation for bus 26, illustrating the base method's inability to fully reflect contribution-based fairness. Nevertheless, the contributing ECs still benefit from participating in the coalition, with final costs lower than their individual operation costs, preserving the principle of individual rationality.

Case 2 - ECs at buses 15, 26 and 56: Next, we relocate the EC at bus 40 to bus 56 to study the impact of location on the cost saving. In this case, the computed cost savings are non-zero for all three ECs, as shown in Fig. 9, which reflects that each EC contributes non-negligibly to the coalition's performance. The result above highlights that the Shapley allocation captures the locational impacts of DER placements and ensures a fair cost distribution.

Scalability: To investigate computational scalability, we expand the study to six ECs at buses 17, 18, 26, 27, 39,

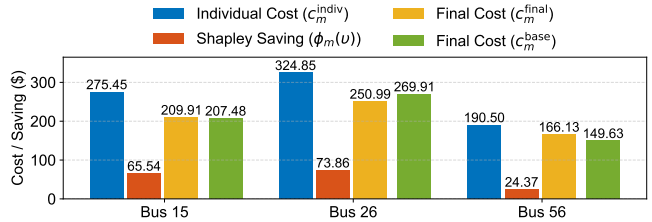

Fig. 9. Individual cost, Shapley-based savings, and final cost for each community bus in the IEEE 69-bus system (ECs at buses 15, 26, 56), under Shapley and Base allocation methods.

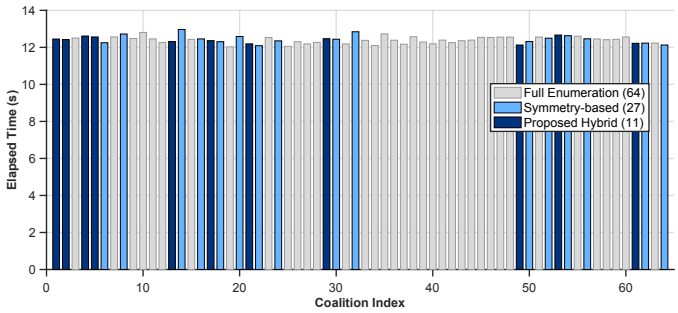

Fig. 10. Elapsed time for all 64 coalitions (Full Enumeration). Sampled coalitions correspond to the 27 symmetry-based (light blue) and the 11 proposed hybrid (navy blue) scenarios.

and 40, resulting in $2^6 = 64$ coalitions. Evaluating the costs for all the coalitions is not computationally scalable, therefore we propose a "signature-based approximation" which aims to reduce the number of coalitions that must be evaluated to obtain same results as when all the coalitions are evaluated.

The proposed approximation leverages two physical insights. *First*, we cluster the ECs with identical DER configurations and electrical proximity; for instance, a coalition including bus 17 produces nearly the same impact as one with bus 18, meaning $\{17, 26, 39\}$ can be represented by the same signature as $\{18, 26, 39\}$. Applying this criteria alone reduces the 64 original coalitions to $3^3 = 27$ unique group-level signatures, since each cluster contains two ECs and can contribute 0, 1, or 2 ECs to a coalition. *Second*, ECs located in different feeders can be considered as independent. So, in our case, the ECs at buses 17, 18, 26, and 27 and ECs at buses 39 and 40 can be treated separately, assuming that their is no influence among these two sets of ECs. Combining this feeder separation with the clustering described above results in $3^2 = 9$ signatures for the first feeder and 3 for the second; removing the duplicated empty coalition yields a total of $9+3-1 = 11$ unique signatures. Thus, only 11 representative signatures are required instead of all 64 coalitions (Fig. 10).

We validate the proposed approximation by comparing the Shapely costs when all the coalitions are considered with the costs when only a reduced number of coalitions are considered. We compare three different versions which are (i) when only symmetry criteria is used for the coalition reduction, (ii) feeder-based criteria is used, and (iii) hybrid when both (i) and (ii) are used. Table IV provides a numerical comparison. As shown in this table, the cost obtained by the hybrid method is quite close to the case when full set of coalitions are simulated. Furthermore, Table V summarizes the solver calls and elapsed times for each method. As can be seen, the hybrid method is most computationally efficient.

TABLE IV
COMPARISON OF SHAPLEY SAVINGS ACROSS DIFFERENT METHODS ($)
(IEEE 69-BUS).

Bus	Full Enum.	Sym.-only	Branch-only	Proposed Hybrid
17	80.606	80.713	80.597	**80.737**
18	80.633	80.713	80.624	**80.737**
26	102.910	103.100	102.894	**103.994**
27	103.480	103.100	103.469	**103.994**
39	0.012	0.012	-0.003	**-0.003**
40	0.012	0.012	-0.003	**-0.003**
Total	**367.653**	**367.650**	**367.579**	**369.454**

TABLE V
COMPUTATIONAL EFFICIENCY ANALYSIS (IEEE 69-BUS SYSTEM).

Execution	Method	Solver Calls (N)	Time (s)
Sequential	Full Enumeration (Base)	64	719
	Symmetry-based only	27	304
	Branch-based only	19	215
	Proposed Hybrid (S+B)	**11**	**126**
Parallel	Full Enumeration	64	107
	Proposed Hybrid (S+B)	**11**	**14.9**

3) IEEE 123-bus system

To further evaluate the scalability and robustness of the proposed scheme, we apply the framework to the IEEE 123-bus test system. We consider two cases as described below.

Robustness of signature-based approximation: We consider six ECs at buses 4, 6, 55, 59, 90, and 94 located in different branches of the network. The six ECs include both identical and non-identical DER capacities to reflect diverse sizing scenarios. We assume that ECs at buses 4 and 6 are identical in size, forming a symmetric pair, whereas bus 55 has a 20% larger capacity than bus 59, while bus 94 has 50% more DER capacity than bus 90. This is intentionally done to test whether the signature-based method, which has a symmetry assumption, works when we have asymmetric DERs.

We address this case as follows. First, we treat the asymmetric pair (e.g., buses 55 and 59) as symmetric during coalition sampling and then distribute the final cost under Shapley allocation linearly with DER size. This uses the rationale that communities with similar setups tend to have costs (or benefits) that are roughly proportional to their installed DER capacities. The results in Table VI show that the approximation produces Shapely savings with reasonable accuracies.

Scalability: To further assess computational scalability, we consider a case with 30 communities leading to $2^{30} \approx 1.07 \times 10^9$ coalitions. The communities are assumed to be distributed across five weakly coupled branches of the IEEE 123-bus feeder, with 2, 4, 6, 8, and 10 communities per branch, respectively. Within each branch, communities with similar DER configurations are assumed to be paired in a two-by-two manner to form symmetric groups, enabling a signature-based approximation that aggregates coalitions with similar structural impacts. Table VII reports the resulting computation time for the IEEE 123-bus system. While exact (full) Shapley value computation based on full enumeration is infeasible for 30 communities, the proposed hybrid signature-based method requires evaluating only 359 coalitions and simulates within approximately 20 minutes on a standard laptop.

TABLE VI
COMPARISON OF EXACT AND APPROXIMATED FINAL COSTS (SHAPLEY)
(IEEE 123-BUS SYSTEM).

Bus	Exact ($)	Approx. ($)	Abs. Error ($)	Rel. Error (%)
4	12.12	11.72	0.40	3.30
6	12.12	11.72	0.40	3.30
55	41.34	41.99	0.65	1.57
59	50.17	50.39	0.22	0.44
90	24.97	24.46	0.51	2.04
94	36.24	36.69	0.45	1.24
Total	176.95	176.95	–	–

TABLE VII
SCALABILITY COMPARISON BETWEEN FULL ENUMERATION AND THE PROPOSED HYBRID METHOD (IEEE 123-BUS SYSTEM).

Method	Solver Calls N	Time (s)
Full Enumeration	2^{30} ($\approx 1.07 \times 10^9$)	*Intractable*
Parallel Proposed Hybrid	359	1120

We emphasize that cost allocation is performed *ex-post*. This allows the proposed method to employ an approximate allocation for intermediate analysis, while higher-fidelity allocations can be conducted offline. Moreover, the total settlement amount across all ECs remains nearly identical to that of the full enumeration case, with only small error as demonstrated above. As a result, the proposed approximation is a practical and scalable solution without significantly degraded accuracy.

V. CONCLUSIONS

This work has addressed the problem of fair cost allocation in energy communities, where the DERs across multiple communities jointly influence DLMPs and overall dispatch outcomes. The key challenge is ensuring that operating cost savings from centralized scheduling are distributed fairly across communities, accounting for their contributions to system-level performance. To this end, a bilevel optimization framework was proposed from the perspectives of the CEA and the DSO. The bilevel problem was reformulated into a tractable single-level MILP using KKT conditions and strong duality. To ensure fair cost distribution among communities, the Shapley value was applied to quantify each community's marginal contribution to system-wide cost savings.

Simulation results demonstrated that the proposed Shapley-based allocation method fairly reflects the locational impact and resource contribution of each community. Furthermore, sensitivity studies showed that communities hosting larger BESS units receive higher cost savings, highlighting the incentive compatibility of the proposed approach. We also developed a coalition reduction approach, a signature-based method to enhance the computational scalability of the Shapely-value based method that enables running the proposed scheme for a large number of energy communities.

APPENDIX A
DERIVATION OF THE LINEAR SINGLE-LEVEL OBJECTIVE

This appendix derives (10) from (2) using the strong duality of the lower-level DSO problem. Since the DSO problem is linear, its dual objective depends only on the constant right-hand sides of the primal constraints. In the present formulation,

nonzero right-hand-side constants appear only in: (i) the bus injections $p_{l,t}^{\text{inj}}, q_{l,t}^{\text{inj}}$, and (ii) the upper and lower bounds on voltages and generator outputs.

Accordingly, the dual objective at time t can be written as

$$\theta_t(\cdot) = -\sum_{l\in\mathcal{N}} \lambda_{l,t}^{p}\, p_{l,t}^{\text{inj}} - \sum_{l\in\mathcal{N}} \lambda_{l,t}^{q}\, q_{l,t}^{\text{inj}} + \sum_{l\in\mathcal{N}} \left(\overline{\lambda}_{l,t}^{gp}\overline{p}_l^{g} - \underline{\lambda}_{l,t}^{gp}\underline{p}_l^{g}\right) + \sum_{l\in\mathcal{N}} \left(\overline{\lambda}_{l,t}^{gq}\overline{q}_l^{g} - \underline{\lambda}_{l,t}^{gq}\underline{q}_l^{g}\right) + \sum_{l\in\mathcal{N}} \left(\overline{\lambda}_{l,t}^{u}(v^{\max})^2 - \underline{\lambda}_{l,t}^{u}(v^{\min})^2\right), \tag{A.1}$$

where $\lambda_{l,t}^{p}$ and $\lambda_{l,t}^{q}$ are the dual variables of the active/reactive power balances (5c)–(5f), and $\underline{\lambda}_{l,t}^{gp}$, $\overline{\lambda}_{l,t}^{gp}$, $\underline{\lambda}_{l,t}^{gq}$, $\overline{\lambda}_{l,t}^{gq}$, $\underline{\lambda}_{l,t}^{u}$, $\overline{\lambda}_{l,t}^{u}$ are the dual variables associated with the generation and voltage bounds. Using the injection definitions, $p_{l,t}^{\text{inj}} = -p_{l,t}^{c}$ for $l \in \mathcal{N}_c$ and $p_{l,t}^{\text{inj}} = -\hat{P}_{l,t}^{\text{load}}$ otherwise, and similarly $q_{l,t}^{\text{inj}} = -q_{l,t}^{c}$ for $l \in \mathcal{N}_c$ and $q_{l,t}^{\text{inj}} = -\hat{Q}_{l,t}^{\text{load}}$ otherwise, the first two terms of (A.1) become

$$-\sum_{l\in\mathcal{N}} \lambda_{l,t}^{p}\, p_{l,t}^{\text{inj}} = \sum_{l\in\mathcal{N}_c} \lambda_{l,t}^{p} p_{l,t}^{c} + \sum_{l\in\mathcal{N}\setminus\mathcal{N}_c} \lambda_{l,t}^{p}\hat{P}_{l,t}^{\text{load}}, \tag{A.2}$$

$$-\sum_{l\in\mathcal{N}} \lambda_{l,t}^{q}\, q_{l,t}^{\text{inj}} = \sum_{l\in\mathcal{N}_c} \lambda_{l,t}^{q} q_{l,t}^{c} + \sum_{l\in\mathcal{N}\setminus\mathcal{N}_c} \lambda_{l,t}^{q}\hat{Q}_{l,t}^{\text{load}}. \tag{A.3}$$

Equations (A.2)–(A.3) show that the DLMP payment terms in (2) appear explicitly within the dual objective. Since strong duality holds, $f_t^{\text{DSO}} = \theta_t$ at optimality. Substituting the dual expression of θ_t into (2) and rearranging terms yields the linear single-level objective (10).

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